

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the different systems units (CGS, FPS, MKS and SI) used in food engineering and the importance of unit conversion?
- Q.24 Define pasteurization and explain its purpose in food preservation. Describe the working of High-Temperature Short-Time (HTST) method.
- Q.25 Describe the construction and operation of a plate heat exchanger (PHE) and where it is used in food industry.

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Roll No.

4th Sem/ Food Technology Subject : Principles of Food Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The SI system of units is a widely used system in food engineering. What does SI stand for?
- a) Standard Integration
 - b) System of Industry
 - c) System International
 - d) Scientific Information
- Q.2 The study of the flow and deformation of food materials is known as:
- a) Thermodynamics
 - b) Physical properties
 - c) Rheological properties
 - d) Mass transfer
- Q.3 In fluid dynamics, the Reynolds number is used to predict:
- a) Fluid temperature b) Fluid color
 - c) Flow characteristics (Laminar or turbulent)
 - d) Fluid density

- Q.4 Which mode of heat transfer involves the movement of fluid molecules?
 a) Conduction b) Convection
 c) Radiation d) Diffusion
- Q.5 Evaporators that use a series of heating effects to remove water efficiently are called:
 a) Pan type evaporators
 b) Forced circulation evaporators
 c) Falling film evaporations
 d) Multiple effect evaporators
- Q.6 The primary goal of pasteurization is to:
 a) Kill all microorganisms including spores
 b) Destroy pathogenic microbes and reduce spoilage organisms
 c) Increase the temperature to boiling point
 d) Enhance the flavour of the food

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 What are the two systems of units mentioned in the syllabus?
- Q.8 Name three physical properties of food.
- Q.9 Name a type of positive displacement pump used in the food industry.

- Q.10 The process of removing water by boiling the product in a vessel is achieved by what equipment?
- Q.11 Which equipment uses high temperature and pressure for sterilization?
- Q.12 Which equipment produces steam for processing operations?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Define rheological properties of food materials.
- Q.14 Define and briefly explain the importance of thermal conductivity.
- Q.15 Write short note on manometer.
- Q.16 Write short note on positive displacement pumps.
- Q.17 Define modes of heat transfer with the help of examples.
- Q.18 What is heat exchanger? Name any three types of heat exchangers used in food industry.
- Q.19 Make neat and clean diagram of plate heat exchanger and label it.
- Q.20 What is evaporator? Describe the working of batch type evaporator.
- Q.21 Write short note on autoclave.
- Q.22 Describe the working of drum drier with the help of diagram.